Sumit Goel Email: sumitgoel58@gmail.com | Website: goelsumit.com

Teaching Statement

I view teaching as an opportunity to share with students some of the most elegant and insightful ideas in economics. I also find it deeply rewarding because it encourages me to think about concepts in more fundamental ways, learn from students' perspectives, and continuously refine my own understanding.

For these reasons, I have eagerly taken up opportunities to teach. At NYU Abu Dhabi, even though postdocs are not required to teach, I taught Intermediate Microeconomics for three consecutive years. More recently, at Ashoka University, I have been teaching Microeconomics and Algorithmic Economics. I designed the latter as an elective course introducing computational perspectives on economic design problems, allowing me to bring my interdisciplinary interests into the classroom. These experiences have been invaluable in shaping my teaching philosophy.

A key element of my teaching philosophy is to establish an overarching theme for the course and reinforce how individual topics contribute to that theme. In Microeconomics, I introduce the subject as a study of how the institutions under which economic agents interact shape the allocation and use of scarce resources. Following a somewhat non-standard structure, I begin with the impossibility results of Arrow and Gibbard–Satterthwaite to highlight the challenge of designing institutions that work universally well. I then introduce the market mechanism and demonstrate its desirable properties through the welfare theorems, first in the simpler context of the Shapley–Scarf housing economy and later in the classical exchange economy. I was particularly encouraged when a student remarked that the course felt like a story rather than a collection of topics. I believe that helping students situate individual concepts within a broader framework enables them to retain ideas more effectively and develop a deeper understanding of the subject.

I strive to make my classes interactive and to arrive at ideas collectively. For instance, before defining Pareto efficiency, I present different allocations to students and ask for their opinions on whether each represents a desirable redistribution of resources. Such discussions naturally lead to the formal definition. Similarly, in my Algorithmic Economics course, after demonstrating that envy-free allocations need not exist when goods are indivisible, I ask students how they think we should proceed and spend time exploring their proposals together before introducing envy-freeness up to one good (EF1) as a natural relaxation. These discussions have sometimes become so lively that I have faced the delightful challenge of getting students to settle down so that we could continue the lecture and examine their conjectures. I find these moments particularly gratifying, as they reflect the curiosity and independent thinking that I hope to cultivate in my students. Such interactions also keep me intellectually engaged and excited to teach.

I try to use relatable examples and analogies when introducing key concepts. While guest lecturing on game theory for a behavioral economics course at NYUAD, I introduced extensive-form games using the ultimatum game. To clarify that the second player's strategy must specify a response for every possible offer, not just the one observed, I asked students to imagine writing an algorithm that would play the game on their behalf. Analogies like this, I believe, serve as simple yet effective heuristics for checking one's understanding.

I welcome feedback and adapt my course content to the diverse needs and backgrounds of my students. When I first taught Intermediate Microeconomics, I placed strong emphasis on precision and generality. Upon receiving feedback about the challenges students were facing, I realized the importance of distinguishing between the level of generality needed to establish a result and the level needed to convey its central insight. I subsequently revised my approach to place greater emphasis on intuition and motivating examples while maintaining mathematical rigor. This led to a more positive experience for both the students and myself. I believe that responsiveness, flexibility, and sensitivity to students' varied backgrounds are essential to effective teaching.

Beyond my experience as an instructor, my time as a teaching assistant at Caltech was formative. I worked closely with several outstanding faculty members and assisted across nine quarters in six courses: Econometrics, Introduction to Economics, Introduction to Finance, Game Theory, Theory of Value, and Algorithmic Economics. This exposure allowed me to observe a wide range of teaching styles and philosophies and to gain familiarity with effective instructional practices. In addition, I have contributed to Econschool, an initiative that mentors undergraduate students in India preparing for graduate studies in economics. As part of this effort, I taught a course on Mathematics for Economists and developed the website econschool.in, which hosts various learning resources.

My teaching experience, together with my broad research interests and interdisciplinary background in economics and computer science, equips me to teach a wide range of courses. At the undergraduate level, I am well prepared to teach core economics courses and would particularly enjoy offering electives in microeconomics or at the intersection of economics and computer science. At the graduate level, I am especially interested in teaching the first-year microeconomic theory sequence and advanced courses in Market Design, Mechanism Design, and Algorithmic Economics. I also look forward to mentoring students at both the undergraduate and graduate levels and advising their research in economic theory.